Protocol placement.

IP → Internet layer (routing b/w Network

TCP/UDP → Transport layer (reliable vs fast communication

HTTP/HTTPS → Application layer (web comm-

DNS → Applicat^n layer (domain → IP resolution

* Day 15 -

DNS- How names become IPs

- What happen when you type google.com in browser?

→ When you type a domain like google.com your browser ask a DNS resolver to translate the domain into an IP address

The resolver check chached records, queries authoritative DNS servers if needed & return the IP so your browser can connect to the server.

DNS Record types.

- A - map a domain to an IP4 address.
- AAAA - map a domain to an IP6 address
- CNAME - Alis for another domain name
- MX - mail exchange server for handling emails.
- NS - Name servers authoritative for for the domain.

(2) IP addressing :-

IPv4 address -

IPv4 is bit address written in dotted notation eg. 192.168.1.10. it consists of four octests ranging from 0-255.

Public IP - accessible on internet

Private IP - used inside local network, not routable on the internet.

(3) CIDR & subnetting

- /24 mean in 192.168.1.0/24?

The /24 indicates that the first 24 bits are the N/W part of the address, leaving 8 bits for host address.

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usable hosts per subnet :

- /24 → 256 total IPs → 254 usable hosts
- /16 → 65,536 total IPs → 65,534 usable ports
- /28 → 16 total IPs → 14 usable ports.

Why subnet -
subnet divides the network into smaller segments to improve efficiency, security & magageability.

(4) Ports.
A port is logical endpoint on a device that allows different service to use the network simultaneously.
Port let a single IP address host multiple services.

Port	Service
22	SSH
80	HTTP
443	HTTPS
53	DNS
3306	MySQL
6379	Redis
27017	MongoDB

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